Learning Guide 6

1.2A Basic steps of a scientific investigation

1.2D Methods for doing scientific investigations (Fair test)

Objectives:

Knowledge: State the variable to be changed, variable to be kept constant and variable to be compared.

Skill: Construct a table to identify the important variables in a fair test.

Attitude: Build up the concept of fair testing.

(I) **Underline** the **vowels** (a, e, i, o, u) and **break up the words** into **syllables** (only for the first word).

(II) Check the dictionary/textbook and write the **Chinese meaning** of the words in the brackets provided.

(III) Review the keywords and prepare for **dictation**.

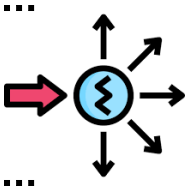
Dictionary
(with pronunciation)

★Q1a. **Keywords (I): Scientific investigation**

Scientific investigation	(*noun phrase*)	(科學探究)	**Diagrams**
(a) observation	(*noun*)	(觀察)	
(b) hypothesis	(*noun*)	(假說)	
(c) result	(*noun*)	(結果)	
(d) conclusion	(*noun*)	(結論)	

Hints: 觀察、假說、結論、結果、科學探究

★Q1b. **Keywords (II): Fair test**

Fair test	(*noun phrase*)	(公平測試)	**Diagrams**
(a) variable	(*noun*)	(變項)	
(b) changed	(*adj.*)	(改變)	
(c) constant	(*adj.*)	(不變)	
(d) compared	(*verb*)	(比較)	

Hints: 公平測試、改變、比較、不變、變項

Preparation Questions

(Read textbook P.14−15 and answer the following questions)

(Check the Chinese meaning of the words highlighted before lesson)

★Q2. There are **FIVE** basic steps for scientific investigation:

(1) Making observations and asking questions.

(2) Doing background research.

(3) Proposing a hypothesis.

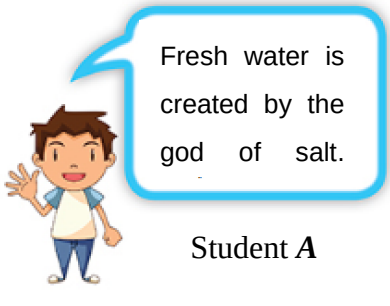
(4) Designing and conducting experiments.

(5) Analyzing results and drawing a conclusion.

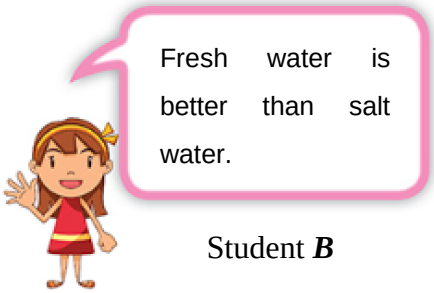
★Q3. A hypothesis is a reasonable answer to the question. It must be testable.

Q4. Below shows three hypothesis suggested from three students, ***A***, ***B*** and ***C***.

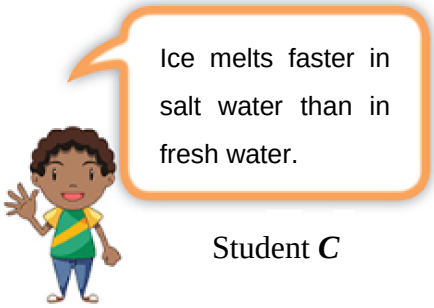




Student ***A***



Student ***B***



Student ***C***

Which student gives the best hypothesis? Explain your answer.

Student ***C*** gives the best hypothesis. Because it can be tested.

Student ***A*** gives a hypothesis which cannot be tested.

Student ***B*** gives a hypothesis which is subjective.

Language Support:

We use the following word phrase patterns to describe the variables in fair test.

e.g.	Volume	of	water
	(*attributes*)	of	(*noun*)

Q5. Make use of following words to form three variables that may appear in fair test.

attributes: size, volume, speed, height, weight, temperature, length, shape

noun: water, Tom, ball, bottle, salt, apple, car

Variable 1: Size of ball

Variable 2: Height of Tom

Variable 3: Temperature of water (*Accept other reasonable answers*)

Q6. The student conducts an experiment to investigate the factors that may affect the time for melting a ball of ice in water. Suggest some of the factors you found may be useful.

(1) *e.g. temperature of water*

(2) Size of the ice ball

(3) Shape of the ice ball

(4) Temperature of the air

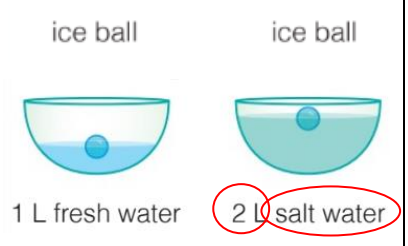
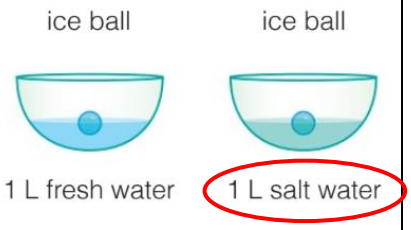
The factors listed here are called variables.

Q7. To find out how other variables affect the time, we can vary one variable each time while keeping **others the same**. In this way, we have done a **fair test**.

Q8. The following table shows three experiments.

(a) Circle the difference(s) in the experimental set-up.

(b) Decide whether the test is a fair test or not.

	Experiment *A*	**Experiment *B***	**Experiment *C***
Circle the difference in the set up			
How many difference(s)?	2	2	1
Is this experiment a fair test?	This (*is* / *is not*) a fair test	This (*is* / *is not*) a fair test	This (*is* / *is not*) a fair test

Q9. Finally, the variables in the experiment ***C*** can be tabulated as follows:

Variable to be changed	**Variables to be kept constant**	**Variable to be compared**
• types of water	• sizes of ice • shapes of ice • volumes of water etc	• time of ice melting

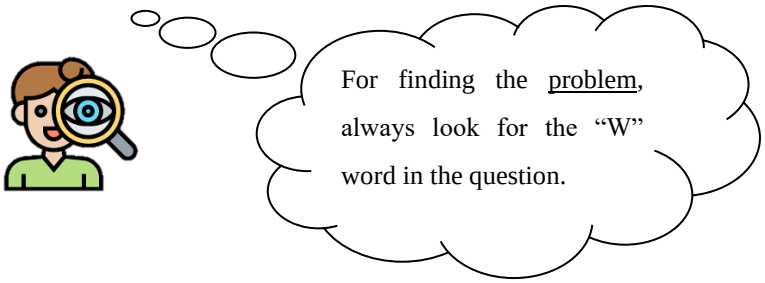
[Skill corner – Teacher's Demonstration]

Q10. *Case study 1*: Simon puts some salt into a glass of water. He wants to know whether the volume of water will affect the amount of salt dissolved. He thinks that more salt will dissolve in a larger volume of water. To test his hypothesis, he carries out a scientific investigation.



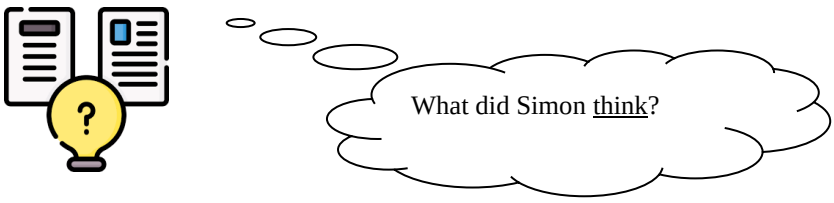
(a) What is the problem to be investigated?

Whether the volume of water will affect the amount of salt dissolved.



(b) What is Simon's hypothesis?

More salt will dissolve in a larger volume of water.



(c) Complete the following table.

Variable to be changed	Variables to be kept constant	Variable to be compared
• volume of water	• temperature of water • size of the cup • size of the salt	• maximum amount of salt that can dissolve in water

Hints:	volume of water	temperature of water	size of the cup
	maximum amount of salt that can dissolve in water		size of the salt

Q11. *Case study 2*: Susan wants to know which brand of washing powder (brand ***A*** or brand ***B***) has a better cleaning effect. She thinks that washing powder of brand ***A*** has a higher ability to remove soya sauce stain than that of brand ***B***. To test her hypothesis, she carries out a scientific investigation.



For finding the problem, always look for the "W" word in the question.

(a) What is the problem to be investigated?

Which brand of washing powder has a better cleaning effect.

(b) What is Susan's hypothesis?

What did Susan think?

Washing powder of brand ***A*** has a higher ability to remove soya sauce stain than that of brand ***B***.

(c) Complete the following table.

Variable to be changed	Variables to be kept constant	Variable to be compared
• brand of washing powder	• temperature of water • size of the stain • amount of washing powder • volume of water • method of washing	• time of cleaning the stain

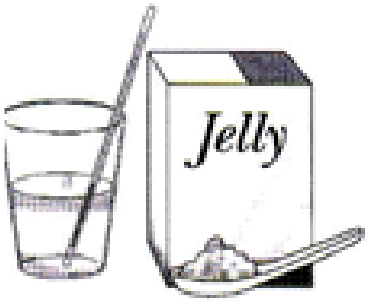
Hints:	temperature of water	size of the stain	volume of water
	brand of washing powder	time of cleaning	method of washing

Experiment: Dissolving jelly powder with water of different temperatures.

Preparations:

Chemicals	Materials / Apparatus	Safety precaution
♦ 80 ℃ water ♦ 25 ℃ water ♦ Jelly powder (1 bag)	♦ 100 mL beaker × 2 ♦ Glass rod × 2 ♦ Spatula × 2 ♦ Stopwatch	Be careful of hot water

Q12. *Case study 3*: You are now going to do a scientific investigation on whether hot water dissolves jelly powder faster than cold water. You think that using hot (*hot / cold*) water will dissolve jelly powder faster than using cold (*hot / cold*) water.



(a) What is the problem to be investigated?

Whether hot water dissolves more jelly powder than cold water.

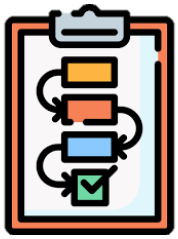
(b) What is your hypothesis?

Hot water will dissolve jelly powder faster than using cold water.

(c) Complete the following table.

Variable to be changed	Variables to be kept constant	Variable to be compared
• temperature of water	• size of the cup • time of stirring • brand of jelly powder	• time needed to dissolve all the jelly powder

Procedure



1. Label one beaker as "**hot water**" and the other as "**cold water**".
2. Measure about 80 mL of 80 ℃ water and pour it into the "hot water" beaker.
3. Measure about 80 mL of 25 ℃ water and pour it into the "cold water" beaker.

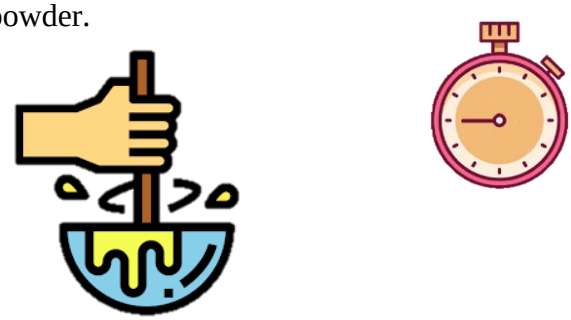


hot water cold water

4. Use the spatula to measure an equal amount of jelly powder from the bag and transfer it to the beaker labeled "hot water". Start the stopwatch after adding all the powder.



5. Stir the mixture using the glass rod until the jelly powder is completely dissolved. Stop the stopwatch and record the time taken to dissolve all the powder.



6. Repeat **Steps 4–5** using the beaker labelled "cold water".

Results and Discussion

(d) Write down the time needed to dissolve all the jelly powder in hot and cold water respectively.

Temperature	25 °C (*cold water*)	80 °C (*hot water*)
Time (in seconds) needed to dissolve all the jelly powder	(*Depends on the experiment*)	(*Depends on the experiment*)

(e) Which kind of water, hot water or cold water, has a shorter dissolving time? Hot water



Conclusion (*what did this experiment tell you?*)

From the experiment. hot water dissolves jelly powder faster (*faster / slower*) than cold water.

13. After this activity, how does you view towards **scientific investigation** has **CHANGED**?

I used to think	**Now I think**
(*Accept any reasonable answers*)	(*Accept any reasonable answers*)